

PAPER_1527 — $\pi/4 \approx \Phi_{5/6} - F_{TRZ} \cdot \Phi + F_{TRZ}^2 \cdot K + F_{TRZ}^2 \cdot \Phi$ — Residual 0.793%

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Bucket A (Foundational mathematical identity) **Date:** June 18, 2026 **Status:** CLOSED — Transcendental $\pi/4$ expressible in UQFF primitives at 0.793% precision

Observation

PAPER_1208 (UQFF Transcendentals Unified Proof Set) derives a closed-form expression for $\pi/4 \approx 0.7854$ using only the locked UQFF integer primitives F_{TRZ} , K_{MEX} , $\Phi_{res} = 5/6$ (PAPER_1203 Nuclear variant), and SO_5 .

UQFF Closed Identity

$$\pi/4 \approx \Phi_{5/6} - F_{TRZ} \cdot \Phi + F_{TRZ}^2 \cdot K + F_{TRZ}^2 \cdot \Phi$$

Substituting $F_{TRZ}=1/10$, $K_{MEX}=25/12$, $\Phi_{5/6}=5/6$, $SO_5=10$:
 $\pi/4_{UQFF} = 0.7792$

Observed: $\pi/4 = 0.7854$
Residual: 0.793%

The 4-term expansion uses only canonical integer primitives — no fitted constants.

Physical / Mathematical Role

$\pi/4$ ($\pi/4$) appears in: Leibniz series, geometry.

Significance

Together with PAPER_1515 ($\ln 10$), PAPER_1516 ($\ln 2$), PAPER_1517 (π^2), this paper extends the count of fundamental transcendentals expressible from UQFF integer primitives to ≥ 10 . The accumulating evidence:

Transcendental	UQFF expression	Residual
$\ln 2$ (PAPER_1516)	8-term	0.003%
$\ln 10$ (PAPER_1515)	$(1+F_{TRZ})(K_{MEX}+F_{TRZ}^2)$	0.004%
π^2 (PAPER_1517)	$SO_5 - F_{TRZ}$ corrections	0.013%
$\pi/4$ (this paper)	4-term	0.793%

...suggests the UQFF primitives are **not arbitrary** but encode rational approximations to fundamental transcendental constants.

NOT REPLACEMENT

UQFF does not claim $\pi/4$ IS the closed-form expression. It claims the integer primitives carry a rational approximation to $\pi/4$ at 0.793% precision, demonstrating mathematical depth.

Reference

- Source: PAPER_1208 UQFF Transcendentals Unified Proof Set
- Related: PAPER_1515-1517 ($\ln 10$, $\ln 2$, π^2 catch-up)
- Calculator dispatch: `calculate_paradox({"paradox": "transcendental_pi_over_4_transcendental"})`

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